

Direction

2398744-100

REVISION 2.0

Advantage 4D

**Service Introduction Letter
(SIL)**

Revision History

REV	DATE	REASON FOR CHANGE
A	Oct. 30, 2003	First draft
1	Nov. 14, 2003	Editions from training review with OLC CT, PET, and AW support leaders:
2	March 3, 2004	REV 2 for M4: Editions for service & sales install flow & eLicense / Structure – to close M3 feedback issues

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Figure 1 -- Advantage 4D "SERVICE" Installation Sequencing

Step 1. Call or visit site to determine the answer to these questions:
Must all be YES to proceed to STEP 2 !

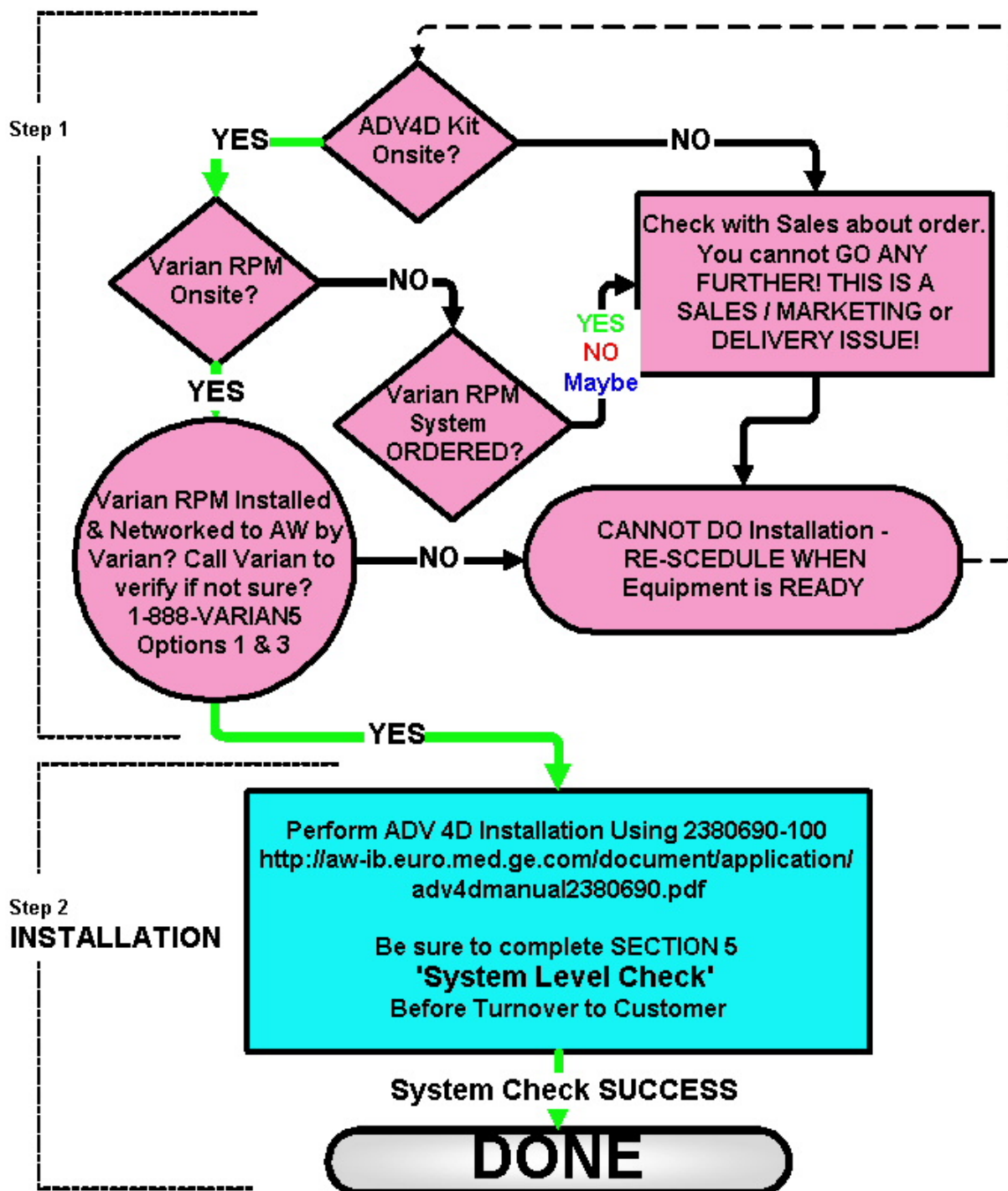


Figure 2 -- Advantage 4D "SALES" Installation Sequencing

Marketing Information Letter; November 18, 2003 Advantage 4D Sales Information

CT / Adv4D Order Information

What's included in the CT B -CAT?

- The Advantage 4D Software for the Advantage (Oncology) Workstation
- Gantry Interface Kit that is mounted inside the CT gantry
- Cables that will connect the CT to the Interface Kit and Interface Kit to the Varian Interface Box

Prerequisites for the CT B -CAT?

- Advantage Workstation 4.1 or higher
- Varian RPM System
- Exact Couch, i.e., flat table top (this is a must to mount the Varian RPM infrared camera)

Ordering areas of Caution

- You must verify the CT System platform to order the compatible Interface Kit

Discovery / Adv4D Order Information

What's included in the PET S -CAT?

- The Advantage 4D Software for the Advantage (Oncology) Workstation
- Cables that will connect the Discovery to the Interface Kit and Interface Kit to the Varian Interface Box
- Discovery Extender Cable Kit

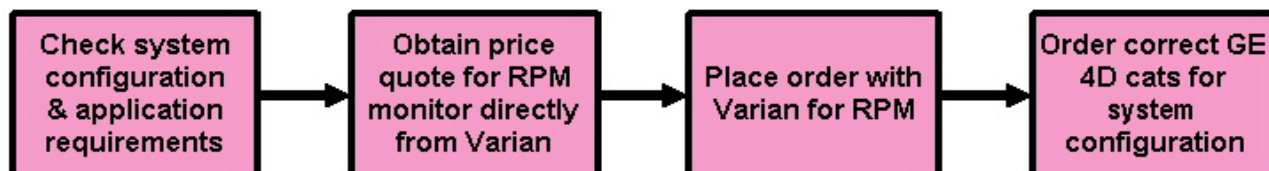
Prerequisites for the PET S -CAT?

- Advantage Workstation 4.1 or higher
- Varian RPM System
- Exact Couch, i.e., flat table top (this is a must to mount the Varian RPM infrared camera)
- Gating Interface Kit (DST: P5851JA, DLS: P5850JA)

Ordering areas of Caution

- Verify Discovery system has Gating Interface Kit installed. If not, order P5851JA for DST: , P5850JA for DLS.

Order Flow



Advantage 4D 'Overview' Information

Advantage 4D CT is a retrospective respiratory gating software application hosting in the Advantage Workstation. CT data is acquired in Cine clusters with multiple acquisitions per longitudinal coordinate. The patient respiratory motion is recorded via the external RPM system. Both the CT acquired images and the respiratory motion data are transferred to the AW as inputs to the Advantage 4D CT Application, which correlates the respiratory phase information to each acquired image. Images then can be saved to the Image Database based on their respiratory phase.

The Advantage 4D hardware has been designed for H1 through H16, H-Power, LightSpeed RT and Future systems. It is shipped ready for immediate installation. In order to be compatible with all of these systems, there are a variety of cable configurations detailed in the installation manual. Be sure to follow the correct cable installation procedure for the system that it is to be installed on. This option has 4 different ways to be connected depending on the gantry type:

- LightSpeed 5.X (LS16 Pro, LS RT Systems)
- LightSpeed 2.X, 3.X, 4.X (LS 16, Ultra, Plus and QX/I Systems)
- LightSpeed 1.X (QX/I Systems)
- Discovery LS and ST Systems

NOTE:

TWO prerequisite service installation issues for this option for SERVICE scheduling / work flow management:

1.) THE VARIAN RPM SYSTEM MUST BE INSTALLED FIRST!

To complete the installation and 'system test' Advantage 4D for turnover, the Varian RPM system MUST be installed, configured, and functional first. The AW - with Advantage 4D software loaded on it alone – can launch only on its demo exam data set. That is, it can only be stand-alone tested. Similarly, the ADV 4D gantry cable(s) & hardware can be installed, and routed (if item 2 below is satisfied), but cannot be system tested for turnover if there is no RPM system to connect it to.

SEE FIGURES 1 & 2 Service & Sales FLOW DIAGRAMS ON PRECEDING PAGES

2.) To install Advantage 4D, the ct gantry system must be equipped with the Cardiac Gating Option. Respiratory and Cardiac Gating can be installed on either side of the gantry for H2 through H16 and HPower systems, but must be installed from the left side of the gantry on H1 systems.

Summary:

- A Varian product is joined with a GE product.
- Both are ordered separately and connected on site.
- Each system will have its own self test.
- Varian supplies an RPM system with a motion phantom, infrared camera, and an interface box.
- GE supplies cables to connect the gantry to the interface box, AW software, and an installation manual.

Pre-Requisites:

GE
Varian

Refer to FIGURE 3 - end of this document - for an overview of the connection flow for the ADV 4D system.

Installation

Advantage 4D will run on AW4.1, and up rev. AW releases.

Advantage 4D has a specific Installation Guide Document: 2380690-100, which is part of the product structure and available on the web at:

<http://aw-ib.euro.med.ge.com/>

http://www.ct.med.ge.com/tech_pubs/www/techpubs/welcome.htm

Advantage 4D software (on the AW) will be protected by an independent “**software license key**” which is a function of the HOSTID/LICENSEID of the workstation. The key should be delivered with the Advantage 4D software package, as an 8 to 10 digit number to type in during the installation, and on a license floppy diskette.

DICOM demo exam(s) will be provided with the software on a separate CD in order to allow startup/run sanity testing of the application, as well as to verify basic stand-alone Advantage 4D functionality.

- Advantage 4D can be pre-loaded onto the hard disk – for future license enabling. This will support potential eFlexTrial implementation.

Diagnostic tools

No diagnostic tools are required. Installation and/or error messages can be seen in the console Window and are logged in the **/var/adm/messages**, and/or **~sdc/logfiles/EasyInstallLog**, and/or the dedicated Advantage 4D error log **~sdc/logfiles/adv4dctlog**

There are specific Advantage 4D configuration files – actually MOTION FILES - that should be saved if the AW platform is going to be re-loaded. Within Advantage 4D there is the option to COPY these files to Hard DISK or to FLOPPY. **Refer to IM; 2380690-100 – chapter 4**

Periodic maintenance

No formal periodic maintenance is required.

However, the AW System Configuration Save should be done immediately before and after ANY software installation - and - at regular service intervals thereafter, in order to save the correct software license key for the Advantage 4D package for restoration after an AW software service intervention.

Corrective maintenance

Consists of saving the Advanatge 4D MOTION FILES to floppy, reloading software and the software license key, and restoring the MOTION FILES from floppy. This can be done by the customer with OLC support if necessary, or the FE with or without OLC support.

Advantage 4D Parts List & Catalog Structure:

To install Advantage 4D, the CT system must already be equipped with the Cardiac Gating Option

CT Cats:

B7840DA	Adv 4D HW for H1/4.1 AW
B7840DB	Adv 4D HW for H2-LS16/4.1 AW
B7840DC	Adv 4D HW for WideBore&HPwr/4.1AW
B7840DD	Adv 4D RespGating HW-WideBore&HPwr
B7840DE	Adv 4D cabling for H1-LS16/4.1AW
B7840DF	Adv 4D cable for WideBore&HPwr

FRU's:

- H1 – H16 Pulmonary Kit 2371490
- Widebore and Hpower Pulmonary Kit 2371491
- Cardiac/Pulmonary Cable 2371401
- Cardiac/Pulmonary Cable 2354761
- PET Cable Extension kit 5103369
- App. SW and Docs on CD-ROM 2393326-2
- App. Demo Exam on CD-ROM 2393325

AW Cats:

B77001FD	Advantage 4D CT SOFTWARE PACKAGE
B77041FD	Advantage 4D CT 'No Charge' Hardware Upgrade

- **B77001FD** - Advantage 4D New purchase
 - o **2373314** **Collector**
 - ✓ **2340775** AW Blank Config. Floppy
 - ✓ **2393327** Advantage 4D Function Supplier Package
 - ✓ **2393276-247** Advantage 4D Customer Release Notes
 - ✓ **2380690-100** Advantage 4D Installation Manual (English)
 - ✓ **2393326-2** Advantage 4D Application CDROM (**FRU**)
 - ✓ **2393325** Advantage 4D Demo Exam CDROM (**FRU**)
 - ✓ **2393284-200** Advantage 4D Quick Step/Users Guide - English
 - -201(French) -206(Spanish) -208(German) -211(Italian)
 - 227(Portuguese) -240(Japanese) -241(Chinese)
- **B77041FD** - Advantage 4D 'No Charge' Hardware Upgrade – To support HARDWARE CHANGE/UPGRADE

PET Cats:

The GOB kit is a prerequisite for Discovery Advantage 4D.

DLS system - **P5850JA**

DST system - **P5851JA**

S9111RG Discovery Advantage 4D

Advantage 4D CABLING PACKAGE PRODUCT STRUCTURE

P5850JD

B7840DE

- 2371490** Pulmonary Kit for H1 thru H16
- 2369602 cable, Gantry to RPM
- 2369603 Cable – Cardiac/Pulmonary
- 2371401 Cable - Cardiac/Pulmonary
- 2371402 Pulmonary Adapter
- 2371403 Cover Plate
- 2380690-100 Installation Instructions

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Advantage 4D CABLE EXTENSION PACKAGE PRODUCT STRUCTURE**P5850JE**

5103369 Cable Extension kit
 2369602 cable, Gantry to RPM
 5103123 Connector, Female, 9 position

Advantage 4D SOFTWARE PACKAGE PRODUCT STRUCTURE**P5850JF**

2373314 Collector
 2393327 Supplier Package
 2380690-100 Installation Manual
 2393326-2 ADV 4D Software CDROM
 2393325 ADV 4D Demo Exam CDROM

Advantage 4D Quick Step Technical Guides

2393284-200 English
 2393284-201 French
 2393284-206 Spanish
 2393284-208 German
 2393284-211 Italian
 2393284-227 Portuguese
 2393284-240 Japanese
 2393284-241 Chinese

Advantage 4D Customer Release Note

2393276-247 ALL Languages

B77001FD

2373314 Collector
 2393327 Supplier Package
 2380690-100 Installation Manual
 2393326-2 ADV 4D Software CDROM
 2393325 ADV 4D Demo Exam CDROM

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 2393284-240 Japanese
 2393284-241 Chinese

Advantage 4D Customer Release Note

2393276-247 ALL Languages

Training

The primary AW technical support people at the OLC will be trained by CSE and CT/PET engineering on the Advantage 4D Installation/Service Manual – 2380690-100 with the use of a powerpoint presentation summarizing the Advantage 4D option. The installation/service document, as well as this SIL will also be available on the AW IB website: <http://aw-ib.euro.med.ge.com/>.

The powerpoint training presentation on the basics of the Advantage 4D option is available at http://supportcentral.ge.com/products/sup_products.asp?prod_id=16107 under the Training link.

Be sure to subscribe to the AW Support Community in Support Central to access this link.

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Documentation

There **WILL** be a separate specific Service (Installation) Manual for Advantage 4D. The Advantage 4D Installation Manual (2380690-100) will support the following service requirements:

- Hardware & Cable interconnections
- AW Advantage 4D software installation
- Overall system test
- Individual stand-alone tests for:
 - AW software system
 - RPM system
 - RPM Interface box

Any Service document as Service Notes, Service Introduction Letter (SIL), etc. will be posted on the AW/IB website (<http://aw-ib.euro.med.ge.com/>), and announced under the OLC e-DISTRIBUTION process for the AW and/or appropriate modality(s).

The latest revisions of the Service Documentation will be available on the AW-IB website and the CT service Engineering Technical Publications Website:

<http://aw-ib.euro.med.ge.com/> - Technical Documentation

http://www.ct.med.ge.com/tech_pubs/www/techpubs/welcome.htm

Tracking process

After installation, complete and send Product Locator information for tracking purposes, and or access:

U.S.

http://egems.gemedicalsystems.com/gib/gib_entry.jsp

Europe

<http://egib.euro.med.ge.com/index.html?b=0.24617015957084543>

Escalation process

- Call OLC – AW Support Team.
- Use **iTRAK system (PQR / PSR)** process to report problems under AW, or appropriate modality.
- Use **CSO** process for escalation **after working with the OLC AW Support team first**.

Figure 3 – General / Basic OVERVIEW of Advantage 4D Respiratory Gating connection/data flow

